P.9 Combinations of Functions: Composite Functions

Name _____ Period ____ Date _____

Pre-Calculus P.9 · Student sheet

I can…

- add, subtract, multiply, and divide functions.
- find and evaluate compositions of functions.
- find the domain of a composition and decompose a function.

Vocabulary · fill in the missing word

Composition of functions — $(f \circ g)(x) =$ _____

Arithmetic combinations — _____

Part 1 · Function machines

Machine A: × 2. Machine B: + 3.

Input	A then B	B then A
1		
2		
3		
x		

Part 2 · Notes and examples (fill in each note, then work the example)

Arithmetic combinations: $(f + g)(x) = f(x) + g(x)$, and so on; for f/g, also need $g(x)$ _____.

Example 1

$f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$. Find $(f + g)(x)$, $(f + g)(3)$, and $(f - g)(2)$.

Example 2

$f(x) = x^2$ and $g(x) = x - 3$. Find $(fg)(x)$ and $\left(\frac{f}{g}\right)(x)$ with its domain.

Composition: $(f \circ g)(x) =$ _____: work from the inside out.

Example 3

$f(x) = x + 2$ and $g(x) = 4 - x^2$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, and $(f \circ g)(-2)$.

Domain of $f \circ g$: x must be in the domain of g, and g(x) must be in the domain of _____.

Example 4

$f(x) = x^2 - 9$ and $g(x) = \sqrt{9 - x^2}$. Find $(f \circ g)(x)$ and its domain.

Example 5

Find f and g so that $h(x) = \dfrac{1}{(x-2)^2} = f(g(x))$.

Part 3 · Partner practice

Partner 1

$f(x) = 3x - 4$ and $g(x) = x^2$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Partner 2

$f(x) = \sqrt{x}$ and $g(x) = x + 7$. Find $(f \circ g)(2)$.